

Entangled fates of upwelling and mesoscale eddies under global warming

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Abstract

Eastern boundary upwelling off Peru sustains strong coastal cooling and high marine productivity, yet its long-term evolution under climate change remains uncertain. A key limitation lies in the fact that coarse-resolution climate models cannot adequately capture the relevant dynamical processes occurring at the narrow coastal scale. Moreover, the neglected mesoscale eddies also shape local wind field responses through coupled air-sea interactions. Here, we analyse EERIE coupled simulations (≈ 5 km ocean, 20 km atmosphere, year 1950-2050) to understand how the Peru offshore upwelling system varies with wind, surface heating, and eddies, and to assess whether this system exhibits increasingly pronounced eddy compensation characteristics in a warming climate.

Project Goal

The goal of this project is to use a high-resolution coupled simulation to (i) establish how Peruvian coastal upwelling and its related mean state evolve from 1950 to 2050, and (ii) identify the mechanisms responsible for these changes. We will quantify the respective roles of wind stress and net surface heat flux in setting the upwelling-driven circulation and upper-ocean structure, and then evaluate how mesoscale eddies modify the net vertical exchange through eddy compensation. In particular, we will assess whether eddy compensation strengthens under continued warming and how this changes the residual upwelling signal and its imprint on coastal SST and thermocline depth.

Hypotheses

(H1) Wind stress sets the direction of upwelling change, while net surface heating and stratification control how strongly that change appears in SST and thermocline depth.

(H2) Mesoscale eddies increasingly compensate the wind-driven overturning over time, so the residual overturning is a better predictor of SST and thermocline trends than wind-based upwelling indices alone. (Building on Thomsen et al. (2021)'s framework)

Planned Diagnostics and Figures

(1) Surface changes (model vs observations; past to future).
Show how coastal winds (**wind stress**), **SST** and **net surface heating** evolve from 1950–2050, and verify the historical period against observations where possible.

(2) Subsurface structure and circulation changes.
Show (x,z) sections (coast-to-offshore vs depth) illustrating how **stratification** and the **subsurface/vertical circulation** associated with upwelling change from past to future.

(3) Eddy role and eddy compensation of upwelling.
Show how mesoscale eddy activity changes (e.g., **EKE**) and quantify **how the eddy-driven circulation offsets the wind-driven upwelling**, highlighting whether eddy compensation strengthens toward the future (**Thomsen et al., 2021** as methodological guidance).

(4) Synthesis (what best explains SST/thermocline trends; optional CMIP comparison).OPTIONAL
Compare wind-only upwelling metrics versus net (residual) upwelling to explain long-term SST and thermocline changes, and optionally contrast with CMIP-class models to illustrate what coarse resolution misses.

- **Thomsen et al. (2021)** Thomsen, S., Capet, X., & Echevin, V. (2021). Competition between baroclinic instability and Ekman transport under varying buoyancy forcing in upwelling systems: An idealized analog to the Southern Ocean. *Journal of Physical Oceanography*, 51(11), 3347–3362.